

14. Quality Management

What is Quality Management

- Quality management is the *act of overseeing all activities and tasks needed to maintain a desired level of excellence.*
- This includes the determination of:
 - › Quality policy
 - › Quality planning
 - › Quality control
 - › Quality improvement.
- It is also referred to as ***Total Quality Management (TQM).***
- Quality management *focuses on long-term goals* through the *implementation of short-term*

Service Quality

- A *comparison* of the *customer expectations* and the *service performance*.
- Measuring the gap between customers expectations and how they perceive the services offered to them.
- *Expected Service (ES)*-word of mouth, past experiences, what customer assumed?
- *Perceived Service (PS)*- what is delivered to him?, what he receives?

Customer Gap (Satisfaction) = Perception – Expectation

- $PS > ES$; *quality surprise*
- $PS = ES$; *satisfactory quality*
- $PS < ES$; *unacceptable*

Cost of Poor Quality (COPQ)

- Defined as *costs that would disappear if systems, processes, and products were perfect.*
- Direct & Indirect poor quality costs.
- Direct-** scrap, rework, changes in design, reinspection of reworked items.
- Indirect-** customer dissatisfaction, loss of reputation.

Total quality management (TQM)

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- Consists of organization-wide efforts to *"install and make permanent climate where employees continuously improve their abilities to provide on demand products and services that customers will find of particular value "*.
- *"Total"* emphasizes that *departments* in addition to production (for example sales and marketing, accounting and finance, engineering and design etc.) *are obligated to improve their operations*.
- *"Management"* emphasizes that *executives are obligated to actively manage quality through funding, training, staffing, and goal setting*.

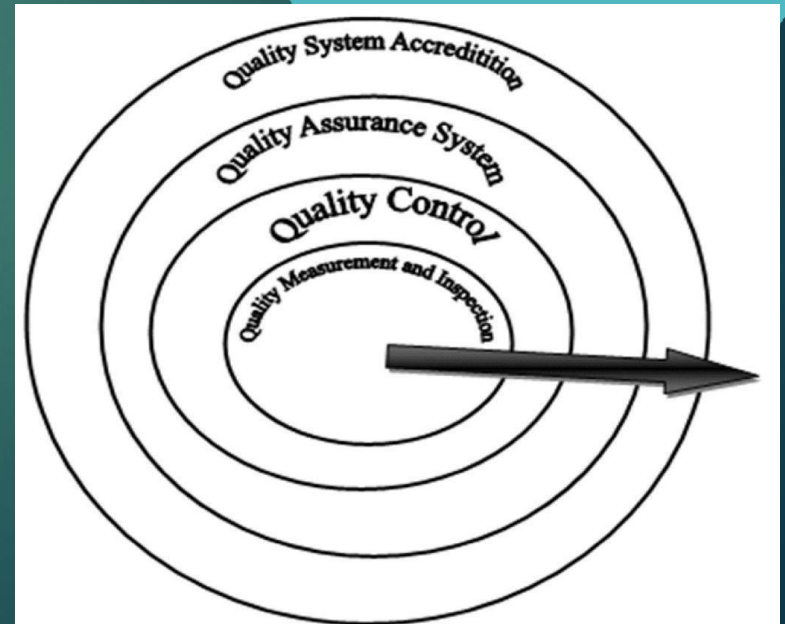
- Total Quality Management (TQM); this basically consists of the Quality Trio.

The Trio is:

- *The Quality Control*
- *The Quality Assurance*
- *The Quality Improvement*

Quality Evolution

- Need for quality *excellence and competition* are the driving source behind *never ending process of improvement*.
- Evolution of quality is divided into four major stages:
 - *Inspection*
 - *Control*
 - *Assurance*
 - *Accreditation*

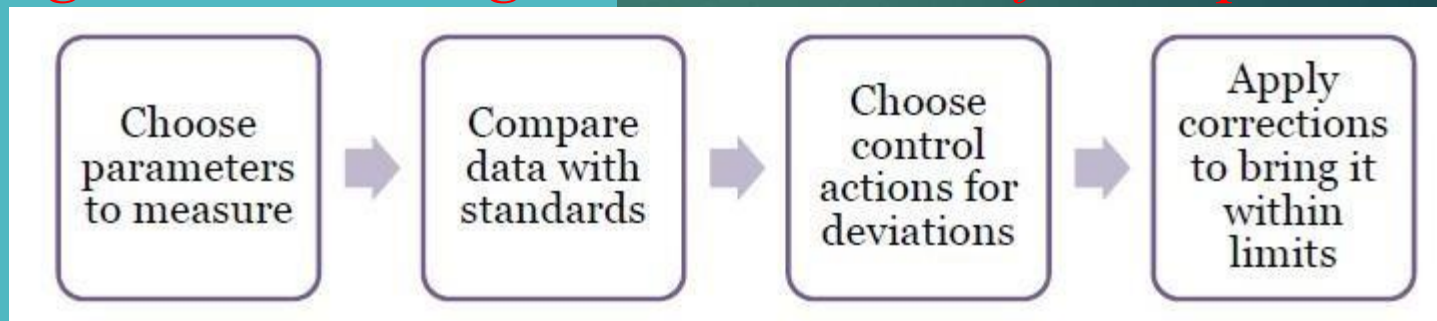


Quality Inspection

- **Quality inspection** are measures aimed at checking, measuring, or testing of one or more product characteristics and to relate the results to the requirements to confirm compliance.
- This task is usually *performed by specialized personnel and does not fall within the responsibility of production workers. Products that don't comply with the specifications are rejected or returned to improve.*
- It is *concerned on post operative or post-production check of products.*
- *Quality inspector only sorts products, and he doesn't let poor quality products to leave the factory.*

Quality Control

- *Tests should not only distinguish bad products from good, but also find the cause of problems.*
- Inspection is done to improve the process by controlling.
- Control actions are based on measured data, recorded on quality control charts
- *Corrective actions are taken as soon as values enter danger zone to bring it within limits defective products*



Quality Assurance System

- ⊙ *Represents an overall system of quality management of an organization, assuring quality excellence through inspection & control actions.*
- ⊙ ISO9000, ISO14000- EMS, SA8000, ISO27000
- ⊙ *Organizations get these certifications after an audit from certifying agency.*
- ⊙ Product standards and system Standards are different.
- ⊙ Standard product under a standard system is the

Quality System Accreditation

- ① *Acknowledgement and recognition of a QMS (Quality Management System) by a national or international body.*
- ① Criteria are broader in scope including checking elements of all aspects of a system. *For example:*
 - *PEC- National*
 - *ABET- International*
 - *Washington Accord- International*

Home Reading

Assignment

- Six Sigma Quality



References

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† Based in part on materials by Ms. Anam Rafique (Former Assistant Professor, Department of Computer Engineering in HITEC University Taxila Cantt).

† Harold Kerzner, “Project Management, A System Approach to Planning, Scheduling & Controlling”, 10th Edition, John Wiley & Sons, Inc, 2012, ISBN: 978-0-470-27870-3.

† Kathryn Bartol, David Martin, Margaret Tein, Graham Matthews “Management, A Pacific Rim Focus” 3rd Edition, McGraw-Hill Companies, Inc, 2002.

† Gerry Johnson & Kevan Scholes, “Exploring Strategy” 8th Edition.

† Related documents from open source, mainly internet.